EN.530.626: Optimal Control for Space Systems
Homework 3
Due Oct 21st 11:59PM

The code for this assignment can once again be found on GitHub at this link and can be downloaded by running `git clone https://github.com/JHU-ACEL/ocss_hw.git` from a terminal window. You can either clone/download the repository out or simply `git pull` if you had cloned out the repository for the last homework. Homework submissions and grading will be managed through Canvas.

Problem	**Points Possible**
Problem 1	20
Problem 2	20
Problem 3	20
Total	**60**

Introduction

This homework will focus on the following technical concepts:

1. Understanding how first-and second-order methods for solving convex quadratic programming problems trade off accuracy with computation time and how this affects downstream performance.
2. Understanding the importance of convex optimization techniques as applied to trajectory generation problems.

To accomplish this, the software development learning goals include,

- Implementing the alternating direction method of multipliers (ADMM) solver for a convex quadratic programming problem.
- Implementing trajectory optimization problem using off-the-shelf trajectory optimization packages such as `cvxpy` and `acados`.

1 Problem 1: Convexity

1.1 Part A [5 Points]

A function $f : \mathbb{R}^{n_x} \to \mathbb{R}$ is linear if-and-only-if the following two properties hold:

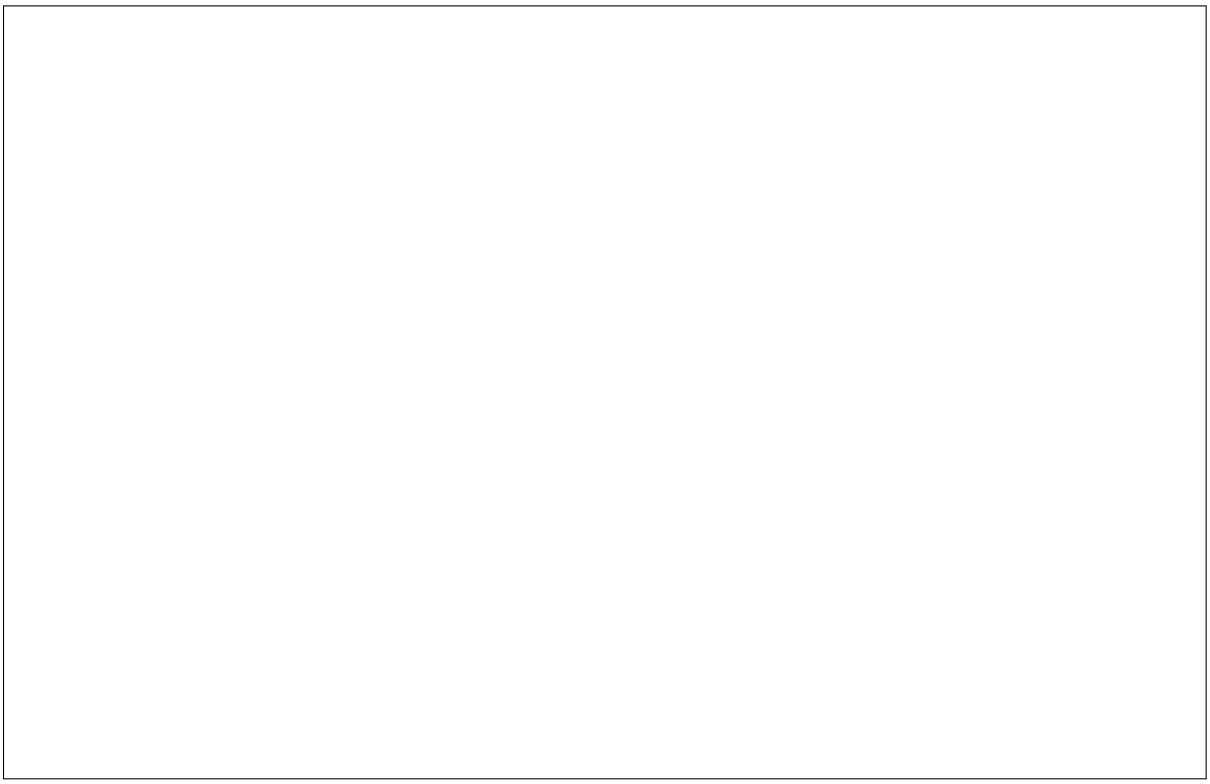
1. For a scalar α, $f(\alpha x) = \alpha f(x)$.
2. For two inputs $x_1, x_2 \in \mathbb{R}^{n_x}$, $f(x_1 + x_2) = f(x_1) + f(x_2)$.

Show that the affine function $f(x) = p^T x + b$ is not a linear function when $b \neq 0$. How can we show under a change-of-variables that this affine function can be made to satisfy the properties of a linear map?

1.2 Part B [5 Points]

A function $f : \mathbb{R}^{n_x} \to \mathbb{R}$ is convex if for all $x, y \in \text{dom} f$ [1] and $\theta \in [0, 1]$, $f(\theta x + (1-\theta)y) \leq \theta f(x) + (1-\theta) f(y)$. Show that the affine function $f(x) = p^T x + b$ is a convex function.

[1] $\text{dom} f$ is the admissible domain of $f(x)$, e.g., for $f(x) = \log x$, $\text{dom} f = \{x > 0\}$.



1.3 Part C [5 Points]

Show that the quadratic function $f(x) = \frac{1}{2}x^T Px + p^T x$, where P is a symmetric matrix, is a convex function if and only if P is positive semidefinite.

1.4 Part D [5 Points]

The Euclidean projection of a point $v \in \mathbb{R}^{n_{\text{ineq}}}$ onto a closed convex set $\mathcal{C}$ is defined as

$$\Pi_{\mathcal{C}}(v) = \arg\min_{z \in \mathcal{C}} \frac{1}{2}\|z - v\|_2^2.$$

For $\mathcal{C} = \{z \mid z \leq 0\}$, show that $\Pi_{\mathcal{C}}(v) = \min(v, 0)$, where the minimum is taken elementwise.

2 Problem 2: Alternating Direction Method of Multipliers

In this problem, we will be revisiting solving the convex quadratic program of the form,

$$\begin{aligned} \underset{x}{\text{minimize}} \ & \frac{1}{2}x^T Px + p^T x \\ \text{subject to } & Ax = b \\ & Gx \leq h \end{aligned}$$

where $x \in \mathbb{R}^n$, $P \in \mathbf{S}^n_+$, and $p \in \mathbb{R}^n$. The linear equality constraints are given by $A \in \mathbb{R}^{n_{\text{eq}} \times n}$ and $b \in \mathbb{R}^{n_{\text{eq}}}$. The linear inequality constraints are $G \in \mathbb{R}^{n_{\text{ineq}} \times n}$ and $h \in \mathbb{R}^{n_{\text{ineq}}}$.

Instead of using the primal-dual interior point method we applied in homework two, we will use the alternating direction method of multipliers (ADMM) algorithm detailed in [1] and encourage you to start this problem by reading Sections 3.1–3.2 and Chapter 5 of this material. Within the ADMM framework, a *splitting* variable $z \in \mathbb{R}^{n_{\text{ineq}}}$ is introduced:

$$\begin{aligned} \underset{x,z}{\text{minimize}} \ & \frac{1}{2}x^T Px + p^T x \\ \text{subject to } & Ax = b \\ & Gx - h = z \\ & z \leq 0, \end{aligned}$$

which can be reformulated as,

$$\begin{aligned} \underset{x,z}{\text{minimize}} \ & \frac{1}{2}x^T Px + p^T x + \mathcal{I}_{\mathcal{C}}(z) \\ \text{subject to } & Ax = b \\ & Gx - h = z, \end{aligned}$$

where $\mathcal{I}_{\mathcal{C}}(z) = \begin{cases} 0 & \text{if } z \leq 0 \\ \infty & \text{otherwise} \end{cases}$ and $\mathcal{C}$ denotes the convex set $z \leq 0$.

The *augmented* Lagrangian $\mathcal{L}_A(x, \lambda, z)$ for this problem is:

$$\mathcal{L}_A(x, \lambda, z) = \frac{1}{2}x^T Px + p^T x + \mathcal{I}_{\mathcal{C}}(z) + \lambda^T(Gx - h - z) + \frac{\rho}{2}\|Gx - h - z\|_2^2,$$

where x satisfies $Ax = b$. Here, $\rho > 0$ and $\lambda \in \mathbb{R}^{n_{\text{ineq}}}$ is the Lagrange multiplier associated with the inequality constraints.

Given some initial iterate (x^k, λ^k, z^k), the ADMM update procedure consists of:

1. $x^{k+1} \leftarrow \underset{Ax=b}{\arg\min} \mathcal{L}_A(x, \lambda^k, z^k)$.
2. $z^{k+1} \leftarrow \Pi_{\mathcal{C}}\left(\underset{z}{\arg\min} \tilde{\mathcal{L}}_A(x^{k+1}, \lambda^k, z)\right)$.
3. $\lambda^{k+1} \leftarrow \lambda^k + \rho \nabla_\lambda \mathcal{L}_A(x^{k+1}, \lambda^k, z^{k+1})$.

Here, $\Pi_{\mathcal{C}}$ in Step 2 is the notation for a projection operator onto the constraint set $\mathcal{C} = \{z | z \leq 0\}$ and we will define $\tilde{\mathcal{L}}_A$ in Section 2.2.

2.1 Part A [4 Points]

Using the necessary conditions of optimality (N.C.O.) for $\mathcal{L}_A(\cdot)$ with respect to x, compute the analytical update rules for x^{k+1}.

Hint: Since x has to satisfy $Ax = b$, for computing x^{k+1}, introduce a Lagrange multiplier y for enforcing $Ax = b$ into the augmented Lagrangian and use equality-constrained Newton method to solve for x^{k+1}.

2.2 Part B [3 Points]

Using the necessary conditions of optimality for $\mathcal{L}_A(\cdot)$ with respect to z, compute the analytical update rules for z^{k+1}.

Hint: The augmented Lagrangian includes a non-differentiable term $\mathcal{I}_{\mathcal{C}}(z)$ — you can ignore this when you compute the N.C.O. with respect to z and the "trick" behind ADMM is to let the $\Pi_{\mathcal{C}}$ take care of the constraint $z \leq 0$. Instead, work with some

$$\tilde{\mathcal{L}}_A(x, \lambda, z) = \frac{1}{2} x^T P x + p^T x + \lambda^T (Gx - h - z) + \frac{\rho}{2} \|Gx - h - z\|_2^2.$$

What would this projection operator $\Pi_{\mathcal{C}}(\cdot)$ be?

2.3 Part C [3 Points]

Compute the gradient of $\mathcal{L}_A(\cdot)$ with respect to λ and provide the update rule for λ^{k+1} for ADMM Step 3.

Hint: The step size for the dual ascent can be simply set to take value ρ (i.e., $\lambda^{k+1} = \lambda^k + \rho \nabla_\lambda \mathcal{L}_A(x^{k+1}, \lambda^k, z^{k+1})$).

2.4 Part D [10 Points]

Putting this altogether — complete the implementation in `admm.py`.

Hint: Unlike with primal-dual interior point methods, our ADMM implementation does not need to start with an initially feasible iterate, (i.e., $Gx^0 \leq h$) and you can instead use a much simpler initial "guess" for the (x^0, λ^0, z^0) tuple.

3 Problem 3: Powered Descent Guidance

In this problem, we will be implementing the lossless convexification approach for rocket powered descent guidance presented in [2]. In this problem, read the instructions provided in `hw3/pdg.py`. There is no additional write-up required for this problem.



Submission Instruction

- Download the marimo notebook for each problem and join them together into a single PDF or HTML file named `hw3.pdf`.
- Compress the `hw3` folder containing your python files. Name this file "hw3.zip".
- Upload both the PDF/HTML file and the zip folder into the canvas assignment "`Homework 3`".

References

[1] S. Boyd, N. Parikh, E. Chu, B. Peleato, and J. Eckstein, "Distributed optimization and statistical learning via the alternating direction method of multipliers," Foundations and Trends in Machine Learning, vol. 3, no. 1, pp. 1–122, 2011.

[2] B. Açıkmeşe, J. M. Carson, and L. Blackmore, "Lossless convexification of nonconvex control bound and pointing constraints of the soft landing optimal control problem," IEEE Transactions on Control Systems Technology, vol. 21, no. 6, pp. 2104–2113, 2013.